

University of Heidelberg



Computational Macroeconomics

Final Assignment (graded)

HAND IN ALONE

hand in in moodle

DEADLINE: 09.08.2026

Teacher
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1 Assignment

This is an **individual** assignment, which means that each course participant should hand in his/her own individual solution. DO NOT work together or copy codes from each other.

Below, a model is presented, that is similar to the one that you worked with over the last weeks. Note however that there are some differences. Further, some tasks are listed that relate to this model, and that you need to execute. Part of your assignment is to do these tasks and hand in both the codes and the results of the tasks.

Additionally, you have to **explain** what your code is doing and *how* it is doing this. Your grade for this assignment will partly be determined by whether your code is correct, and partly by whether you show that you understand your codes well. It is recommended that you do this by adding comments to all your lines of code in your file with codes, explaining in this way, line by line, what your code is doing.

If you write your code in Spyder, you can add comment lines by starting them with `#`. If you write your code in Jupyter notebook, you can add cells of type 'markdown' to write text in your .ipynb file.

If you write your code in spyder in a .py file, please hand in 2 documents:

- a .pdf file that presents the results of the tasks and answers to questions raised in the tasks (a.o. contains plots at all questions where you are asked to create these and contains discussion and interpretation of result where you are asked for this)
- a .py file with codes that I can execute and with comments that explain your code

If you do the assignment in a Jupyter notebook, you can do all the above in one document and hand in only the .ipynb file where I can see all your results and comments and where I can run all your codes.

Please upload these file(s) on moodle in the assignment element (paper with hand symbol) called 'final assignment'. The deadline is **Sunday August 09, 23:59 hours**. Assignments that are handed in after this deadline receive a grade of 5.0.

You can earn a maximum of 50 points for the assignment, and another 50 points for the exam. The bonus points that you have earned during the semester will be added to your score for the final assignment and the exam before your grade is calculated. In order to be able to pass the course you need to have a sufficiently high score for the exam excluding bonus points.

1.1 Model

The equations below present inflation and output in the New Keynesian model which you have studied in the lectures, but with a slight change. The first term

$$Y_t = E_t Y_{t+1} - \frac{1}{\sigma} (i_t - E_t \pi_{t+1}). \quad (1)$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa Y_t + \mu_t, \quad (2)$$

However, the Taylor rule is different than in the lectures and previous assignments: the nominal interest rate now reacts to lagged values of inflation and output rather than to expectations.

$$i_t = \phi_\pi \pi_{t-1} + \phi_y y_{t-1} + \epsilon_t^i, \quad (3)$$

Finally, as you see in the above equations, there are two shocks, one to **inflation** (μ_t) and one to the **nominal interest rate** (ϵ_t^i). The shock to the nominal interest rate is white noise (no auto-correlation and normally distributed with mean zero).

However, the shock to inflation follows an AR(1) process that can be described by

$$\mu_t = \rho_\mu \mu_{t-1} + \epsilon_t^\mu, \quad (4)$$

with ϵ_t^μ a white noise term (random variable with no autocorrelation and normally distributed with mean zero), and ρ_μ the auto-correlation coefficient of the shock μ_t .

Given that past inflation and past output gap show up in the Taylor rule, these variables are state variables. Moreover, the 2 white noise terms are also state variables, and so is the lagged value of the autocorrelated shock μ . The minimum state variable solution will therefore look like this:

$$z_t = C_\mu \mu_{t-1} + C_y y_{t-1} + C_\pi \pi_{t-1} + C_{\epsilon\mu} \epsilon_t^\mu + C_{\epsilon i} \epsilon_t^i \quad (5)$$

For some vectors of coefficients C_μ , C_y , C_π , $C_{\epsilon\mu}$ and $C_{\epsilon i}$.

Unless otherwise stated, use the following values for the parameters in the model

```

1 sigma= 1
2 kappa=0.3
3 beta=0.995
4 phi_pi=1.5
5 phi_y=0.1
6 rho_mu=0.7

```

1.2 Tasks

1. Write the model in the following form:

$$Az_t = ME_t z_{t+1} + Dz_{t-1} + u_t, \quad (6)$$

with the vector

$$Z_t = [\mu_t, y_t, \pi_t, i_t]' \quad (7)$$

and write down the 4x4 matrices A, M and D in your code (Having these matrices correctly in your code would be enough to answer this question, and you do not need to hand in your derivations).

2. In the steady state of the model it holds that $u_t = 0$ and $z_{t-1} = z_t$. Further (assuming that expectations are correct in steady state), it should also hold in steady state that $E_t z_{t+1} = z_t$.

Write a function that has as input a vector of 4 coefficients that can be interpreted as the guessed values of the steady state of the model. In the function, set $E_t z_{t+1}$ and z_{t-1} both equal to this vector of guessed steady state values. Given that you set $u_t = 0$, it is now possible to calculate the right hand side of Equation 6.

After that, your function should calculate the implied z_t vector from Equation 6 and return the differences between this vector and the vector of guessed steady state values that was the input of the function.

3. Use fsolve on the function you have just defined to show that the steady state of the model is (very close to) 0, so that it is indeed correct to not include a constant term in the minimum state variable solution in Equation 5.
4. Define a new function. The function should have four matrices as input arguments. The first three are the matrices A, M and D and the fourth input argument should be an initial guess for the matrix F that contains all coefficients on lagged values in the MSV solution.

Your new function should return the solution for the matrices F and Q that satisfy $z_t = Fz_{t-1} + Qu_t$. That is, define a function that applies the linear time iteration algorithm which you have worked with in Assignment 4.

Hint remember the formulas

$$F_{new} = (A - MF)^{-1}D$$

and

$$Q = (A - MF_{new})^{-1}$$

5. 'Call' the function that you just defined with the matrices you defined in question 1 to obtain the general solution, $z_t = Fz_{t-1} + Qu_t$, of the model of this graded assignment.
6. Save the correct columns of the matrices F and Q as vectors that correspond to the 5 vectors in the minimum state variables solution: C_μ , C_y , C_π , $C_{\epsilon\mu}$ and $C_{\epsilon i}$.
7. Calculate impulse responses to a shock to μ_t of size $\epsilon_0^\mu = 0.01$ in the same way you did in Assignment 4:
 Make a loop of **30 periods**, where you set $\epsilon_t^\mu = 0.01$ in the first period, and $\epsilon_t^\mu = 0$ in all later periods. Further set $\epsilon_t^i = 0$ in all periods.
 In the loop you can then in each period calculate the variables in z_t , using the vectors C_μ , C_y , C_π , $C_{\epsilon\mu}$ and $C_{\epsilon i}$ of the MSV solution.
8. In a similar way, calculate impulse responses to a shock to ϵ_t^i of size $\epsilon_0^i = 0.01$.
9. Make two figures, each with four subfigures. The subfigures should contain the 4 variables in z_t . In the first figure, plot the impulse responses to the shock μ_t and in the second figure plot the impulse responses to the shock ϵ_t^i .

Make sure that

- All lines are thick (e.g. 4pt)
 - Each line in a figure has a different color or linestyle
 - The subfigures fit well in the figure.
10. Interpret the impulse responses that you have plotted. What is the economic intuition for the variables to respond in the way they do? What difference do you notice between responses to the auto-correlated and the not auto-correlated shock? Does the model return immediately to steady state when the shock to the nominal interest rate is over? Why or why not?
 11. Now run a time series simulation of 500 periods of your model, where you draw random values for ϵ_t^μ and ϵ_t^i in each period. Save your resulting time series of Y_t and π_t in a pandas DataFrame with column names that indicate which column corresponds to which variable.
 12. Use the MSV solution (which you have obtained above) to calculate explicitly what rational expectations in any given period about inflation in the next period are. That is, calculate $E_t\pi_{t+1}$. Include this new variable in your pandas DataFrame.

Hint: under rational expectations it is assumed that agents know the current value of shocks, and, therefore, also the current values of endogenous variables (inflation etc). These you can, therefore, assume to be known

when you calculate expectations based on the (1-period-forwarded) MSV solution.

13. Calculate the forecast errors of the inflation expectations that you have just obtained ($\pi_{t+1} - E_t\pi_{t+1}$).

Hint: think about the timing of the columns in the data frame. You may need to 'shift' a column.

14. Run an OLS regression of forecast errors about inflation on the level of inflation in the period that the forecast was made. Or put differently, estimate β (and α) of the following regression model.

$$\pi_{t+1} - E_t\pi_{t+1} = \alpha + \beta\pi_t + e_t \quad (8)$$

How can you interpret your result?